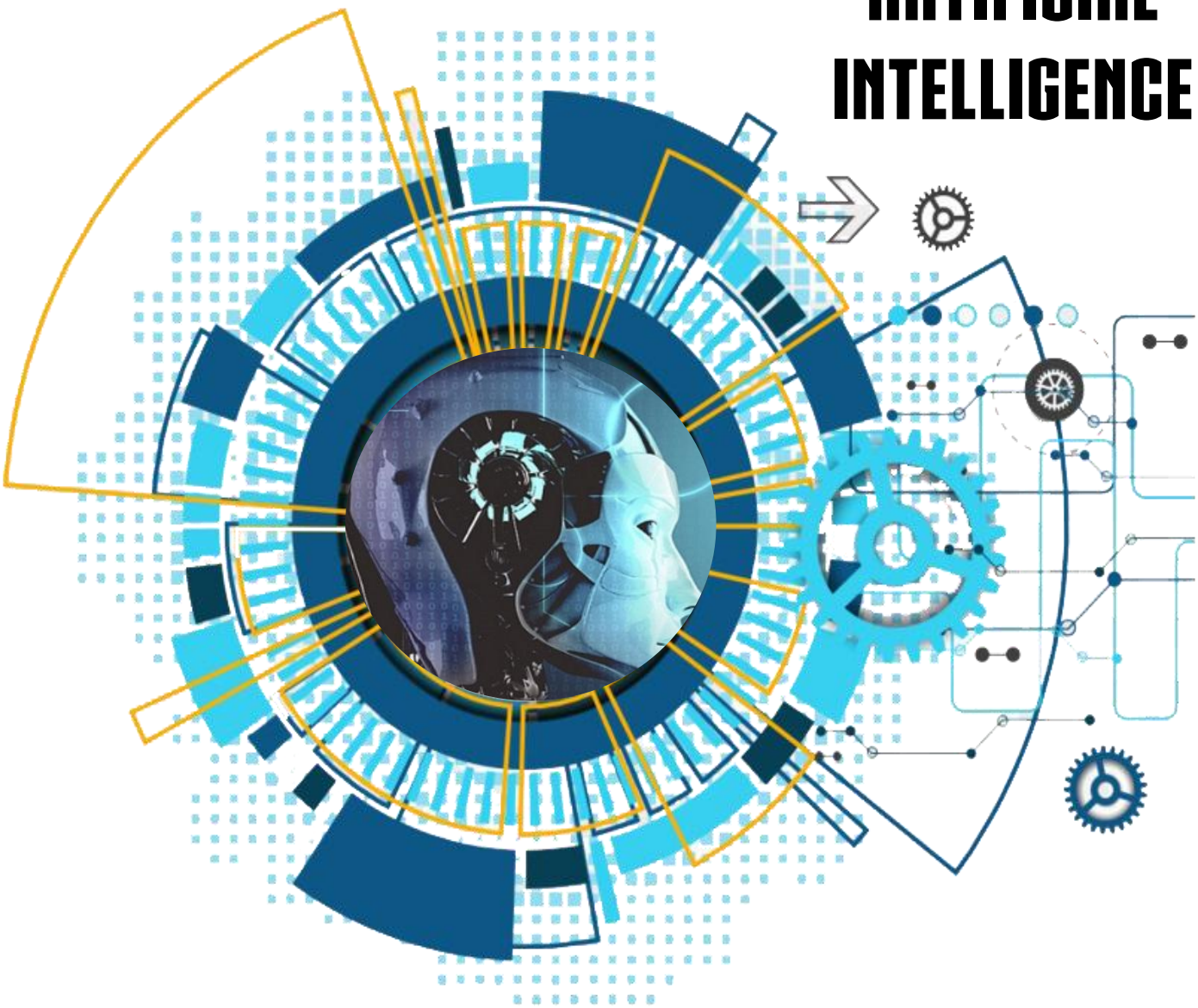
ARTIFICIAL INTELLIGENCE



LEARNING COMPETENCIES

2.1. EXPLAIN THE ETHICS IN ARTIFICIAL INTELLIGENCE.

2.2 IDENTIFY THE TYPES AND TOOLS IN AI SOFTWARE.

2.3. FAMILIARIZE WITH SOME OF THE MOST USED FRAMEWORKS AND TOOLS USED IN AI.

2.4 UNDERSTAND HOW AI WORKS AND THE FIVE ESSENTIAL COMPONENTS THAT CONTRIBUTE TO THE SUCCESS OF ARTIFICIAL INTELLIGENCE.

ACTIVATING PRIOR KNOWLEDGE

1. What is Artificial Intelligence?

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2. What is Artificial Intelligence Ethics?

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3. What are the different types of Artificial Intelligence?

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INTRODUCTION

Artificial intelligence (AI) and robotics are digital technologies that will have significant impact on the development of humanity in the near future. They have raised fundamental questions about what we should do with these systems, what the systems themselves should do, what risks they involve, and how we can control these.

The notion of "artificial intelligence" (AI) is understood broadly as any kind of artificial computational system that shows intelligent behaviour, i.e., complex behaviour that is conducive to reaching goals. In particular, we do not wish to restrict "intelligence" to what would require intelligence if done by *humans*. This means we incorporate a range of machines, including those in "technical AI" that show only limited abilities in learning or reasoning but excel at the automation of particular tasks, as well as machines in "general AI" that aim to create a generally intelligent agent.

In this module, we outline the ethical issues of human use of AI and robotics systems—which means we look at issues that arise with certain uses of the technologies

which would not arise with others, we will also have the opportunity to explore more about the different types of Artificial Intelligence, what are the Artificial Intelligence Tools/Frameworks, what are AI software and types of AI Software's and lastly how do AI works.

DATA INPUT

Artificial Intelligence Ethics

Artificial *Intelligence ethics* is a system of moral principles and techniques intended to inform the development and responsible use of artificial intelligence technology. As AI has become integral to products and services, organizations are starting to develop AI codes of ethics. An AI code of ethics, also called an *AI value platform*, is a policy statement that formally defines the role of artificial intelligence as it applies to the continued development of the human race.

The purpose of an AI code of ethics is to provide stakeholders with guidance when faced with an ethical decision regarding the use of artificial intelligence.

Isaac Asimov, the science fiction writer, foresaw the potential dangers of autonomous AI agents long before their development and created The Three Laws of Robotics as a means of limiting those risks

In Asimov's code of ethics;

1. The first law forbids robots from actively harming humans or allowing harm to come to humans by refusing to act.
2. The second law orders robots to obey humans unless the orders are not in accordance with the first law.
3. The third law orders robots to protect themselves insofar as doing so is in accordance with the first two laws.

Kelly Combs, director at Digital Lighthouse, KPMG said that when developing an AI code of ethics "it's imperative to include clear guidelines on how the technology will be deployed and continuously monitored." These policies should mandate measures that guard against unintended bias in machine-learning algorithms, continuously detect drift in data and algorithms, and track both the provenance of data and the identity of those who train algorithms.

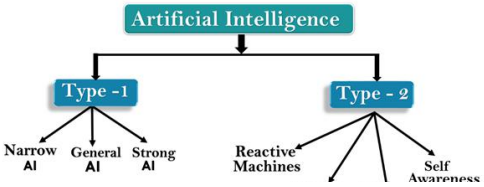
Why are AI ethics important?

AI is a technology designed by humans to replicate, augment or replace human intelligence. Poorly designed projects built on data that is faulty, inadequate or biased can have unintended, potentially harmful, consequences. Moreover, the rapid advancement in algorithmic systems means that in some cases it is not clear to us how the AI reached its conclusions, so we are essentially relying on systems we can't explain to make decisions that could affect society.

An AI ethics framework is important because it shines a light on the risks and benefits of AI tools and establishes guidelines for its responsible use.

Types of Artificial Intelligence

Artificial Intelligence can be divided in various types, there are mainly two types of main categorization which are based on capabilities and based on functionally of AI. Following is flow diagram which explains the types of AI.



AI type-1: Based on Capabilities

Weak AI or Narrow Artificial Intelligence

- Narrow AI is a type of AI which is able to perform a dedicated task with intelligence. The most common and currently available AI is Narrow AI in the world of Artificial Intelligence.
- Narrow AI cannot perform beyond its field or limitations, as it is only trained for one specific task. Hence it is also termed as weak AI. Narrow AI can fail in unpredictable ways if it goes beyond its limits.

  Examples of Narrow AI

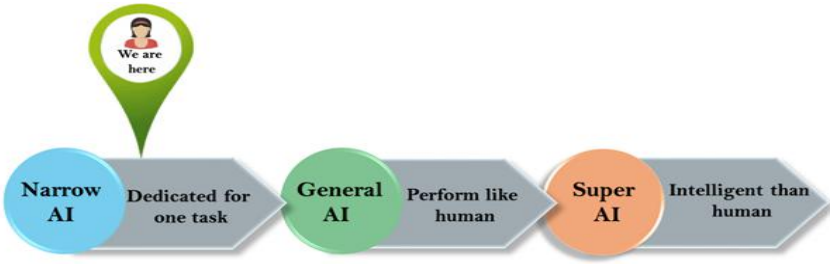
  Playing chess
  Purchasing suggestions on e-commerce site,
  Self-driving cars
  Speech recognition, and
  Image recognition

General Artificial Intelligence

- General AI is a type of intelligence which could perform any intellectual task with efficiency like a human.
- The idea behind the general AI to make such a system which could be smarter and think like a human by its own.

Super Artificial Intelligence

- Super AI is a level of Intelligence of Systems at which machines could surpass human intelligence, and can perform any task better than human with cognitive properties. It is an outcome of general AI.
- Some key characteristics of strong AI include capability include the ability to think, to reason, solve the puzzle, make judgments, plan, learn, and communicate by its own.



Artificial Intelligence type-2: Based on functionality

Reactive Machines

- Purely reactive machines are the most basic types of Artificial Intelligence.
- Such AI systems do not store memories or past experiences for future actions.
- These machines only focus on current scenarios and react on it as per possible best action.

Example of Reactive Machines

IBM's Deep Blue system
Google's AlphaGo

Limited Memory

- Limited memory machines can store past experiences or some data for a short period of time.
- These machines can use stored data for a limited time period only.
- Self-driving cars are one of the best examples of Limited Memory systems. These cars can store recent speed of nearby cars, the distance of other cars, speed limit, and other information to navigate the road.

Theory of Mind

- Theory of Mind AI should understand the human emotions, people, beliefs, and be able to interact socially like humans.
- This type of AI machines is still not developed, but researchers are making lots of efforts and improvement for developing such AI machines.

Self-Awareness

- Self-awareness AI is the future of Artificial Intelligence. These machines will be super intelligent, and will have their own consciousness, sentiments, and self-awareness.
- These machines will be smarter than human mind.

- Self-Awareness AI does not exist in reality still and it is a hypothetical concept.

Artificial Intelligence Tools/Frameworks

Artificial Intelligence (AI) is the talk of the century because each day AI is making the world a better and an easy one. Big names such as Google, Facebook and Amazon are already developing frameworks and tools and contributing them in the form of open-source AI tools. In this section, we are going to see some of the most used frameworks and tools that are used in AI.

- **Caffe in Artificial Intelligence Tools**

  Caffe was developed by Berkeley Vision and Learning Centre and is a deep learning framework that is very popular and widely used among AI engineers and even enterprise users because of its speed. Caffe is capable of processing more than 50 million images all in a single day. The areas where Caffe is extensively utilized are research areas projects, speech, multimedia, and visions.

- **Tensor Flow**

  Tensor flow is an open-source framework that was developed by Google and is used for numerical computation intelligence. It does the computation by using data flow graphs. If we visit the website, https://www.tensorflow.org/, we can see lots of tutorials and learning that anyone can get and start with using tensor flow.

- **Theano in Artificial Intelligence Tools**

  Theano is yet again a very popular open-source library that was developed at the University of Montreal, Quebec, Canada by the LISA group. Theano is similar to tensor flow if we leave apart few differences. While Tensor flow is better at GPU support, data visualization options, Theano supports a wider range of operations than Tensor flow does.

- **Keras in Artificial Intelligence Tools**

  Keras is an open-source neural network library that is programmed in Python language. It has the capability to run on top of other libraries such as Tensor Flow, Theano, etc. It was developed by an engineer at Google, Francois Chollet.

  The way Keras work is – it does not handle any low-level computation, instead, it uses other libraries such as Tensor flow and Theano to do it. So Keras handles high-level API and it compiles model with loss and optimizer functions. If we

visit the website, https://keras.io/, we can see lots of tutorials and learning that anyone can get and start with using Keras.

- **Scikit-Learn in Artificial Intelligence Tools**

  Scikit learn is again an open-source machine learning library that is programmed in python. It was developed by David Cournapeau as a part of the Google Summer of Code project in the year 2007. Scikit learn provides a number of supervised and unsupervised machine learning algorithms that can be used inside of your python program.

  This library is based on Scientific Python and it must be installed before we can start using the sci-kit-learn library.

- **Pytorch in Artificial Intelligence Tools**

  PyTorch is a scientific package that is based on Python and it uses the power of GPU (Graphics processing units). It offers an easy-to-use API and it also provides an excellent platform that offers dynamic computational graphs that can be changed during the run time.

Artificial intelligence software

Artificial intelligence software is "Software that is capable of intelligent behaviour." In creating intelligent software, this involves simulating a number of capabilities, including reasoning, learning, problem solving, perception, and knowledge representation.

Today, Artificial intelligence software is at work in applications such as your smartphone assistant, ATMs that read checks, voice and image recognition software on your favourite social network, and in the software that serves up ads on many of the websites you use. These are just a sample of the growing number of applications of artificial intelligence software that we'll see in the future.

Types of Artificial Intelligence Software

Below are the different types of Artificial Intelligence Software that are currently available. The software comes handy for building and developing smart applications right from scratch.

- **Google Cloud Machine Language** is a software helps to train the system of users. The elements include the Google cloud platform console, gcloud, and REST API. Google Cloud helps in analysing, training, and modifying the user's system. The developed and designed system gets deployed in user infrastructure.

  Google Cloud ML has three components include Google Cloud Platform Console, gcloud, and Rest API to design, analyze and deploy on the user interface. It provides stability and firm support.

- **Azure Machine Learning Studio** is a tool used for deployment of user design as web applications in cloud works on an independent platform and able to use available data sources. It offers solutions to browser-based problems. It is scalable, simple, and easy to use. Here no programming skills are needed and can be integrated with open-source techniques.
- **Tensor Flow** is a numeric computational tool in an open-source system. This is an ML repository for research and development. The solutions can be deployed on CPUs, TPUs, GPUs, clusters of mobile phones, and desktops. It offers good communication support and regular functionalities. Both beginners and experienced can use API for development by the Tensor flow. But it takes time to learn about its complete procedure.
- **H2O AI** is software used for banking, healthcare, insurance, marketing, etc. This is open-source and it allows a user to apply programming languages like R and Python to design systems. Here AutoML feature is included and supports many techniques like gradient boosted machines and deep learning. This program provides a linear platform and executes distributed memory structure**.**
- **Cortana** is a virtual assistant and runs many tasks simultaneously by setting a reminder and giving solutions to the problem. It operates on Windows, iOS, Android, and Xbox OS. It can also execute a simple task from switching off AC to ordering a cake. It uses Bing search engines and its supporting languages apart from English include Portuguese, Chinese, Italian, and Spanish. It is operated via voice control to save time. But here the main disadvantage is some Fitbit scenarios available only in the US.
- **IBM Watson** operates like a Q and A session that offers services to SUSE Linux Servers in the framework of Apache Hadoop. It gathers knowledge from small information and for application development it uses API. It is a robust system that makes smarter business.
- **Sales force Einstein** works as a smart Customer Relationship Management system that is used for marketing, sales, commerce, and analytics and provides more awareness about the available opportunities that capture and process the data by adding new entities. It operates based on history by prioritization. It suggests the best products. Image recognition gives deeper insights into specific products. It requires no data preparation and management of systems.

- **Infosys Nia** promotes enterprises by solving complex tasks and helps in improving the system to empower the business. The core components are the data platform, Knowledge platform, and automation platform. It provides a conversational interface and automation for programming tasks. Cognitive and predictive automation lies on the automation platform. Knowledge platform is used for capturing, processing and reusing the data. The Data platform serves the machine learning platform and data analytics.

- **Amazon Alexa** is similar to Cortana that can understand English, German, French, Italian and Japanese. It is a cloud-based service and can be integrated with existing products using Alexa Voice Service. It can be connected with millions of devices and Bluetooth devices like entertaining systems, camera, lights, etc.

- **Google Assistant** is a virtual assistant by Google and can be used on smart home devices and mobiles. Android, iOS, and KaiOS are supporting operating systems. Many languages are available like English, German, Japanese, Italian, Dutch, Portuguese, Russian, etc. It is used as a dual way conversation. It can do all the services like setting alarms, display Google account information, event scheduling and also can do hardware settings to the devices and used to recognize objects, songs and learn visual information. It can be installed on a car, phone, speaker, watch, laptop.